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Randomness of finite sequences

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Problem: Is there a link between randomness and compressibility?

Compressible objects are definitely not random. For instance, π 's decimals look random, but once one knows where they come from, they can be retrieved deterministically.

We could verify that biased binary sequences, in which one of the digits is sparse, are compressible. To get an explicit code, we may represent the relative positions of the minority digits. The following binary string:

1011111111111111111111111111011111111111111011110111

can be represented by the numbers 1 27 16 5, which indicates that they are zeros at locations 1, 1+27, 1+27+16 and 1+27+16+5. This binary string of length 53 would require 65 bits to be represented using the doubling-length code previously considered, as we should prefix it with 111101110101 to signal its length.

If we code the numbers 1 27 16 5 instead, we get (spaces inserted just for legibility):

101 11001011011 11001010000 1110101 00 110111

The two bits 00 are used to signal the end of the code; the next number indicates how many trailing ones there are.

This makes a total of 40 bits instead of 65 (more concise coding methods can of course be found).

Sparse coding

0/1 point (ungraded)

Consider a binary string of length 10 000 that contains only 1% ones.
What is the compression that the above method would achieve?

☐ about 63%

☒ about 67%

☐ about 87%

☐ about 98%

✗

Answer

Incorrect:

Ones are separated by one hundred zeros on average. These intervals require about 13 bits with the doubling-length code (for instance, 100 is coded as 111101100100). The overall code length is about $13 \times 100 = 1300$ bits, instead of $28 + 10000$ bits. So the correct answer is 87%;.

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You have used 1 of 1 attempt

✗ Incorrect (0/1 point)

Question: If non-random sequences are compressible, what about the converse?

Answer: Yes. Random sequences are incompressible.

This answer may seem surprising at first glance. As we saw, talking about incompressibility for a finite sequence without mentioning the compressing machine is impossible, as you can always find a machine for which the sequence has complexity 1 (or 3 for a prefix machine).

One may nevertheless call 'random' sequences that your favourite machine U (or your favourite programming language) is unable to compress. A great achievement of Andrei Kolmogorov is to have understood that.



definition,
then it passes all the frequency tests.
But the converse is not true, as illustrated by the Champernowne constant.
It is easy to see that even if π looks random, it is by no means random, as its Kolmogorov complexity is very small.
Conclusion:
Randomness is not a probabilistic notion;

A binary sequence of length n can be considered **random for a machine U** if it cannot be compressed by more than c bits, where the tolerance value c is small compared to n .

Randomness of finite sequences

$$C_U(x) > n - c$$

Randomness = incompressibility

Non random number

1/1 point (ungraded)

Find a good reason to consider that the number 19683 is *not* random.

- ☒ 19683 = $(3^{**}3)^{**}3$
- ☐ 19681 is a prime number, just below 19683
- ☐ 19683 occurs at location 71434 in π 's decimal development
- ☐ 19683 is the end of Maria Martinez's phone number



Answer

Correct:

A non-random number is compressible. $(3^{**}3)^{**}3$ is a concise description of 19683. Other characterizations of 19683 do not spare information.

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✓ Correct (1/1 point)

Problem: How does the algorithmic definition of randomness escape the Berry paradox?

The Berry paradox, when applied to randomness, seems to ruin any possibility of defining it. The Berry paradox sounds like the paradox of interesting numbers.

The paradox of interesting numbers

All integers are "interesting!"
You don't buy it?
So consider the smallest "non interesting" integer.
This number turns out to be terrifically interesting!

You probably found a solution to this paradox. It has to do with the false hypothesis that 'interesting' can be a predicate (i.e. a Boolean function, either true or false).

Here is another version of the paradox.

The Berry paradox

The Berry paradox: Considers "the smallest positive integer *not* definable in fewer than 100 words".
But... this defining phrase has less than one hundred words!

Can you solve the Berry paradox? This time, the fuzziness of the property cannot be invoked. The solution to the paradox is the same as in the case of Borel's undefinability-of-randomness paradox. According to Gregory Chaitin (2005, p. 106), Emile Borel stated a version of the Berry paradox that applies to the definition of randomness.

Borel's undefinability-of-randomness paradox

The smallest integer that satisfies the criterion of randomness is unique and remarkable and therefore, not random at all!

Borel's_undefinability-of-randomness_paradox

1/1 point (ungraded)

Borel's "undefinability of randomness" paradox does not apply to the algorithmic definition of randomness, because...

- ☐ No integer is really random.
- ☐ Compressibility is a fuzzy, gradual notion.
- ☒ Compressibility is based on complexity, which is not computable.
- ☐ If a number n is compressible, then $n - 1$ is compressible as well.



Answer

Correct:

The paradox relies on the (false) assumption that the randomness criterion be computable. But as we saw, complexity is not computable. The fact of being random cannot be used to define a number, as it cannot be checked.

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✓ Correct (1/1 point)

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